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M. STUPAVSKY, C. P. GRAVENOR and D.T.A. SYMONS

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Paleomagnetism and Magnetic Fabric of the Leaside and Sunnybrook Tills near Toronto, Ontario

M. STUPAVSKY
C. P. GRAVENOR
D.T.A. SYMONS
Department of Geology, University of Windsor, Windsor, Ontario N9B 3P4, Canada

ABSTRACT

The late Wisconsin Leaside till and the early Wisconsin Sunnybrook till found near Toronto, Ontario, were examined to determine their paleomagnetic characteristics. More than 300 specimens were tested; they showed that both tills retain a stable primary remanent magnetization of detrital origin (DRM) after alternating field demagnetization. The site mean remanence directions give pole positions of lat 72.6° N., long 167.7° W. (8.5°, 11.1°) for the Leaside till and of lat 71.8° N., long 83.8° E. (8.6°, 13.7°) for the Sunnybrook till. These two remanence direction populations are statistically distinguishable with more than 99 percent confidence. Tills can be characterized by their virtual geomagnetic pole positions.

The location of the Leaside till pole provides field evidence to suggest that the DRM of glacial tills is not subject to inclination errors. Anisotropies of magnetic susceptibility indicate that the Leaside till was deposited from southwesterly flowing ice, and the Sunnybrook till, from northwesterly flowing ice. *Key words: stratigraphy, glaciation, paleomagnetism.*

INTRODUCTION

A paleomagnetic study of the late Wisconsin Port Stanley till (Gravenor and others, 1973) confirmed that when magnetic particles are released from the base of a glacier they orient in the Earth's magnetic field to produce a detrital remanent magnetization (DRM) in the resulting till. In the same study, it was shown that the Port Stanley till has a stable remanent magnetization (RM_c) after alternating field (AF) demagnetization. The RM_c of the till is uniform in direction for at least 10 m vertically (Gravenor and others, 1973) and 100 km laterally (Stupavsky and others, 1974); this indicates that the till sheet can be characterized by its virtual geomagnetic pole position (VGP). If other till sheets can be characterized by their VGPs, then this property would be useful in correlating and dating tills.

In some clastic sediments, the RM_c direction is biased toward shallow inclinations by as much as 25° (Irving, 1964, p. 30) so that the calculated VGP is in error and on the far side of the true magnetic pole (relative to the sampling site). Two approaches can be taken to study the RM_c of tills and the possibility of inclination error. First, laboratory studies can be made (Stupavsky and Gravenor, 1974). Second, more till sheets can be studied paleomagnetically to see if they give unique directions and especially to see if some give near-sided VGPs. This study of two till sheets follows the second approach.

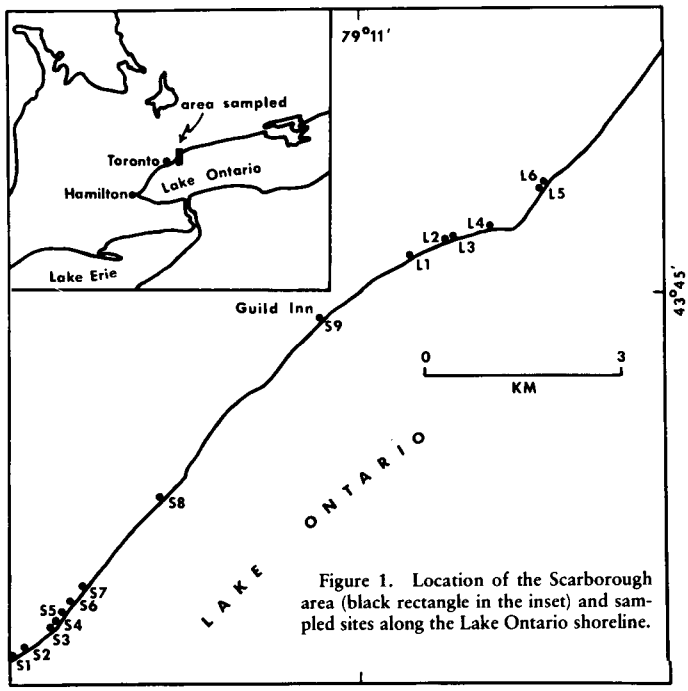


Figure 1. Location of the Scarborough area (black rectangle in the inset) and sampled sites along the Lake Ontario shoreline.

DESCRIPTION OF TILLS

The Leaside and Sunnybrook tills were sampled at 6 and 9 sites, respectively, along the Scarborough Bluffs on the north shore of Lake Ontario just east of Toronto, Ontario (Fig. 1). The Pleistocene stratigraphic sequence exposed in the bluffs as described by Karrow (1967) is shown in Figure 2. The following is a description of the two till sheets as given by Karrow (1967):

The composition of the early Wisconsin Sunnybrook till ranges from fine-grained silt to silty clay with a low pebble content. An average granulometric analysis of the till shows that it contains 45 percent clay, 37 percent silt, and 18 percent sand. It is massive in character and forms vertical cliffs along the Lake Ontario shoreline. The late Wisconsin Leaside till is split into two units — an Upper Leaside and a Lower Leaside (Dreimanis and Karrow, 1972). The Leaside till sampled is from the Lower Leaside but is the uppermost till in the area sampled. It is a silty-sand till containing an average of 20 percent clay, 33 percent silt, and 47 percent sand.

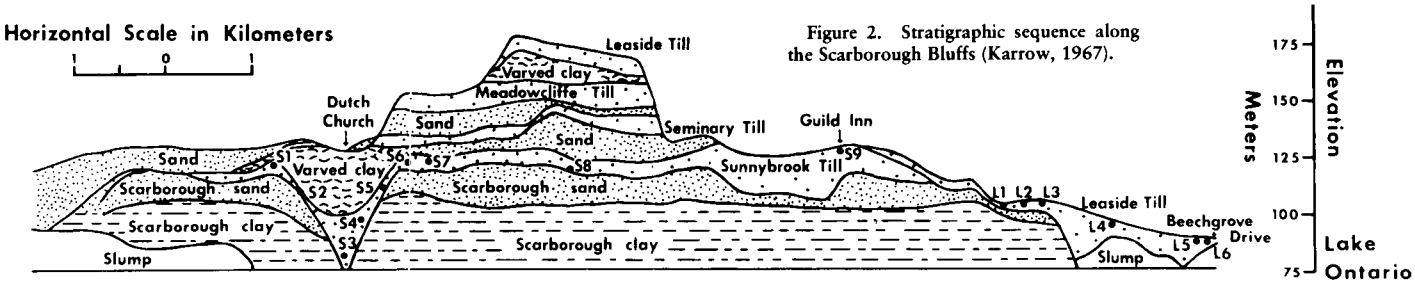


Figure 2. Stratigraphic sequence along the Scarborough Bluffs (Karrow, 1967).

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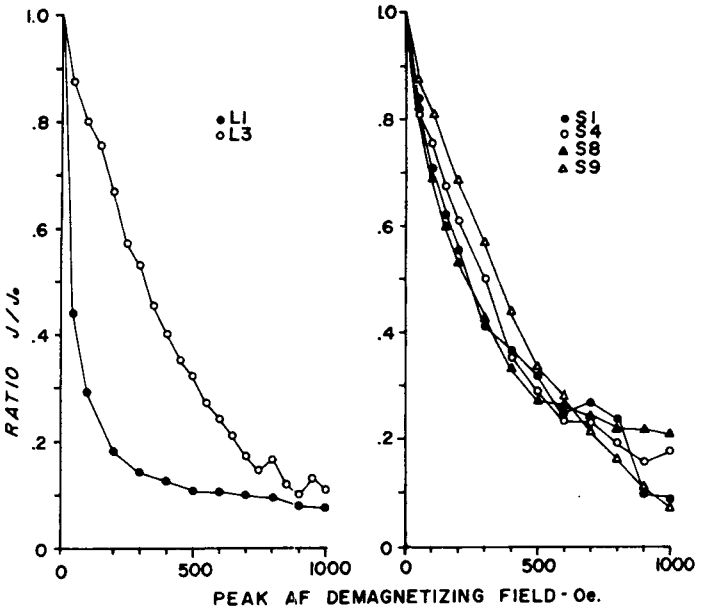


Figure 3. AF demagnetization curves for 2 specimens of the Leaside till and 4 specimens of the Sunnybrook till plotting the remanence intensity ratio J/J_0 against the demagnetizing field. The J_0 values for L1 and L3 are 1.12×10^{-5} and 4.54×10^{-5} emu/cc; the J_0 values for S1, S4, S8, and S9 are 8.03×10^{-6}, 1.51×10^{-5}, 2.33×10^{-5}, and 2.32×10^{-5} emu/cc, respectively.

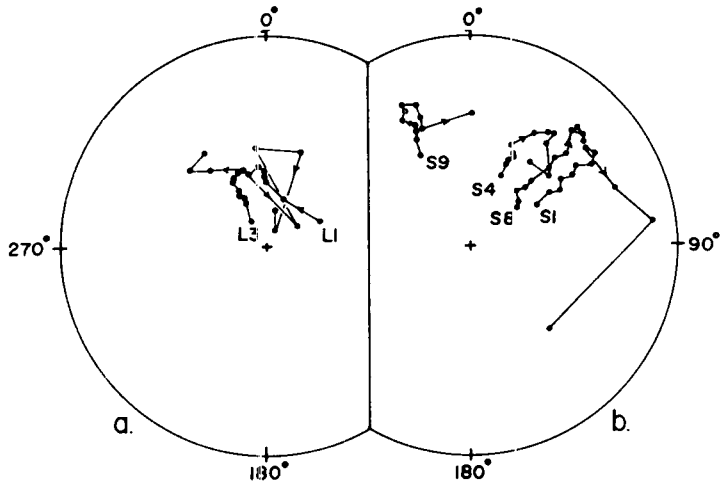


Figure 4. The change in remanent magnetization direction on AF step demagnetization for two specimens of Leaside (a) and 4 specimens of Sunnybrook (b) tills. From the NRM direction, the remanence goes progressively to the direction after 50, 100, 150, 200, 300, 400, 500, 600, 700, 800, and 900 Oe.

SAMPLING AND MEASUREMENT OF REMANENCE

Between 2 and 7 blocks of till measuring 10 × 10 × 15 cm were cut at each site from the cliffs in a vertical sequence at about 10-cm intervals using a portable rock saw. An average of 5 specimens were cored from each block. The natural remanent magnetization (NRM) of one specimen per site was measured on a Schonstedt SSM–1A spinner magnetometer. The remanence of these specimens was remeasured after AF demagnetization by a Schonstedt GSD–1 demagnetizer with step increments of 50 or 100 Oe up to a maximum of 1,000 Oe.

Of the 6 specimens of Leaside till, 4 exhibited a nearly linear decrease in remanence intensity on AF demagnetization (for example, specimen marked L1 in Fig. 3). The intensity of the remaining 2 specimens decreased very rapidly up to 100 Oe (for example, specimen L3 in Fig. 3). The median destructive field or AF intensity required to reduce the NRM intensity by one-half was 254 ± 156 Oe. All 9 of the Sunnybrook till specimens exhibited nearly linear decrease in RM_c intensity (Fig. 3). They required a median destructive field of 295 ± 47 Oe. The remanence directions of specimens from both tills behaved similarly in that they moved away from the present Earth's magnetic field in the initial 50 to 200 Oe of AF demagnetization (Fig. 4) as viscous remanence components (VRM) were removed. The directions then clustered at demagnetizing fields of 400 to 700 Oe, after which they scattered increasingly as anhysteretic remanence components were induced in the specimens by the demagnetization process. In accordance with the results of these cleaning tests, the remaining specimens of the Leaside till were AF demagnetized at either 150 or 200 Oe, and those of the Sunnybrook till, at 300 or 400 Oe. Then their remanence was measured.

TABLE 1. SUMMARY OF REMANENT MAGNETIZATION DATA

Site	Location*	Number of specimens	Number of blocks	Length of resultant vector R	Mean remanence direction			
					Declination (°)	Inclination† (°)	Precision parameter k	α_{95} (°)
Leaside till								
L1	A 1026	22	3	2.983	337	56	1'9	11
L2	A 1027	14	3	2.997	348	66	598	5
L3	A 1027 B	39	7	6.977	346	65	256	4
L4	A 1029	25	6	5.970	325	71	169	5
L5	A 1032	27	3	2.996	328	57	515	5
L6	A 1032 B	11	2	1.974	335	49	38	42
Total		138	24					
Mean by sites (6)				5.9412	336.4	60.6	85	7.3
Sunnybrook till								
S1	A 1007	28	6	5.899	25	57	50	10
S2	A 1006 A	11	2	1.993	23	35	139	21
S3	A 1006 B	18	2	1.997	352	53	352	13
S4	A 1008	20	4	3.986	22	40	213	6
S5	A 1008 A	25	3	2.993	360	32	271	8
S6	A 1009	34	5	4.959	32	49	97	8
S7	A 1009 A	28	4	3.954	341	39	65	12
S8	A 1014	16	4	3.929	4	47	42	14
S9	A 1022	30	4	3.973	342	33	113	9
Total		210	34					
Mean by sites (9)				8.6577	5.8	44.3	24	10.9

* After Karrow (1967).
† Positive inclination indicates down.

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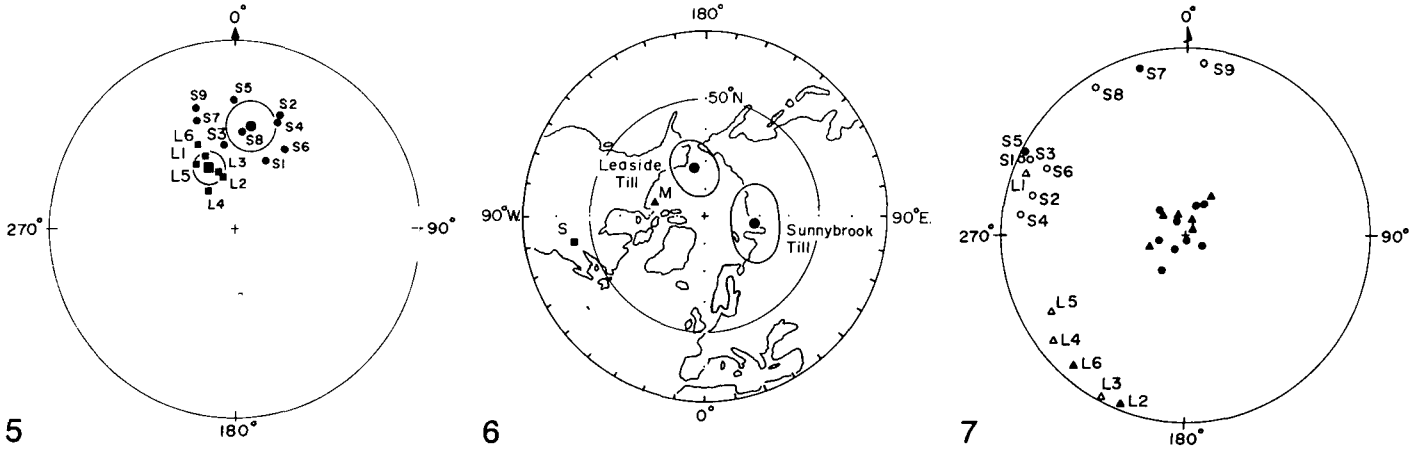


Figure 5. Site mean remanence directions. Circles represent Sunnybrook till; squares, Leaside till. The large solid circle is the mean position for the Sunnybrook till; the large solid square, the mean position for the Leaside till. The large open circles show the 95 percent confidence level. Figure 6. Polar projection of the northern hemisphere showing the VGP positions (solid circle) and the oval of 95 percent confidence for the Leaside and Sunnybrook tills. The location of the Scarborough area is marked S, and the present Earth's magnetic pole is designated M. Figure 7. Anisotropy of magnetic susceptibility data showing site mean directions of maximum and minimum susceptibility. The maximum susceptibility direction of the Leaside and Sunnybrook tills are labeled by the site numbers. Solid symbols indicate positive or downward inclinations on the stereonet; open symbols, negative or upward. The minimum susceptibility directions are represented as solid circles (Sunnybrook) and solid triangles (Leaside) in the center of the stereonet.

Each of the 3 to 10 specimen remanence directions per block were given unit weight to calculate the block mean direction ($\mathbf{RM}_b$). The $\mathbf{RM}_b$ directions were averaged to yield the site mean direction ($\mathbf{RM}_s$; Fig. 5) along with its precision parameter (k) and radius of 95 percent confidence (α_{95}; Fisher, 1953; Table 1). The $\mathbf{RM}_s$ directions were averaged to obtain the mean direction ($\mathbf{RM}_m$) for each till sheet along with its k and α_{95} values.

From the $\mathbf{RM}_m$ directions and the dipole formula, the VGP position for the Leaside till (Fig. 6) is lat 72.6° N., long 167.7° W. with the semiaxes of its oval of 95 percent confidence of 8.5° (δ_p) and 11.1° (δ_m). The VGP for the Sunnybrook till (Fig. 6) is lat 71.8° N., long 83.8° E. ($\delta_p = 8.6°$; $\delta_m = 13.7°$).

DISCUSSION OF REMANENCE RESULTS

The within-site variations in $\mathbf{RM}_b$ directions are small for most sites; this results in high k and small α_{95} values for the mean site directions (Table 1). The between-site to within-site angular variance ratio δ_b^2/δ_w^2 is greater than the theoretical statistic F at the 95 percent confidence level for both the Leaside and Sunnybrook tills (Table 2). Thus, it is likely that the site mean directions are significantly distinct within each till sheet (Larochelle, 1968) despite the inability of the AF cleaning to completely remove the VRM components (Fig. 4). This result agrees with the suggestion of Gravenor and others (1973) that the within-site variations in $\mathbf{RM}_b$ directions reflect variations in the Earth's magnetic field direction during the few decades of time when the till was deposited. Similarly, the between-site variations in the $\mathbf{RM}_s$ directions reflect the variations of a few hundred years in the time of deposition of the till at different sites.

Our findings are consistent in that the Leaside till is thin and was sampled for a short horizontal distance (Figs. 1 and 2); this resulted in a small between-site variation (Fig. 5, Tables 1 and 2). Conversely, the Sunnybrook till is much thicker and was more widely sampled; a larger between-site variation resulted.

The VGP positions for the Leaside and Sunnybrook tills are geographically well separated and have nonintersecting ovals of 95 percent confidence (Fig. 6). This implies that each till sheet may be characterized by its VGP position. It also suggests that secular variation is not being averaged out in the tills; this means that the tills were deposited rapidly within a period of a few hundred years.

When sediment grains settle through a water layer, the detrital remanent magnetization (DRM) of the sediment agrees in declination with the applied magnetic field but is often less in inclination; this results in an "inclination error." Inclination errors of as much as 25° have been reported in laboratory sedimentation experiments, depending on factors such as grain size, grain shape, water-flow velocity, and water depth (Irving, 1964, p. 30). An inclination error of 25° would place the calculated VGP about 35° on the far side of the true magnetic pole relative to the sampling site and hence is cause of concern. Most experiments, however, have been carried out using well-sorted particle sizes (for example, Nagata, 1961) under limited hydrostatic heads, whereas tills are poorly sorted and deposited under potentially large hydrostatic heads. Also, the till particles probably were released in a thin slurry and acquired the DRM by particle rotation after deposition but before consolidation. In this case, experiments show that the DRM would be aligned parallel to the applied field (Irving and Major, 1964). Laboratory experiments carried out by Stupavsky and Gravenor (1974) on artificial till slurries suggest that natural tills would not show an inclination error. In this study, the VGP for the Leaside till is not "far-sided" and therefore provides field evidence to support the hypothesis that the DRM of tills is not subject to inclination error.

MEASUREMENT OF MAGNETIC SUSCEPTIBILITY ANISOTROPY

Measurement of the magnetic susceptibility anisotropy has been used as a rapid method for till-fabric determination (for example, Fuller, 1964). In this study, the anisotropy of 6 to 31 specimens from each site was measured on a Schonstedt SSM–1A magnetometer using a 6-spin method; a toroid transformer bridge (Christie and Symons, 1969) was used to obtain the bulk susceptibility along one axis. For each specimen, the magnitude and direction of the minimum, intermediate, and maximum susceptibility directions were calculated by the method of successive approximations (Nye, 1960). Data from specimens in which the ratio of the magnitude of the maximum to intermediate susceptibility exceeded 1.008 were then averaged for each site (Fig. 7, Table 3). About 10 percent of the specimens had ratios of less than 1.008, which was taken to indicate insufficient anisotropy to be acceptable.

TABLE 2. ANGULAR VARIANCE ANALYSIS

Till sheet	Number of sites B	Number of blocks N	Within-site variance δ_w^2 (radians)^2	Between-site variance δ_b^2 (radians)^2	Ratio of angular variances δ_w^2/δ_b^2	Theoretical statistic F*
Leaside	6	24	0.0232	0.0809	3.49	2.68
Sunnybrook	9	34	0.107	0.344	3.21	2.13

* The statistic $F_{2(B-1),\ 2(N-B),\ 0.05}$ is given at the 95 percent confidence level.

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TABLE 3. ANISOTROPY OF MAGNETIC SUSCEPTIBILITY DATA

Site	Number of specimens	Direction of maximum susceptibility			Direction of minimum susceptibility			Ratio of axial magnitudes	
		Declination (°)	Inclination* (°)	α_{95} (°)	Declination (°)	Inclination* (°)	α_{95} (°)	Maximum: intermediate	Minimum: intermediate
Leaside till									
L1	29	293.1	-10.2	12.8	314.6	76.5	5.1	1.036	0.912
L2	9	202.1	3.3	21.9	22.2	82.1	2.8	1.029	0.910
L3	5	208.3	- 0.4	32.3	37.1	84.9	4.1	1.012	0.912
L4	10	232.7	11.0	17.7	34.2	69.9	3.3	1.019	0.910
L5	10	240.2	-20.6	17.1	254.0	74.3	5.3	1.021	0.920
L6	7	221.5	- 9.9	28.6	334.5	80.6	15.1	1.025	0.945
Sunnybrook till									
S1	14	295.9	- 1.1	11.5	32.0	74.8	2.9	1.025	0.925
S2	14	283.9	-16.1	7.7	259.8	78.3	4.2	1.022	0.912
S3	22	296.6	- 3.7	7.6	215.3	71.2	7.2	1.041	0.953
S4	8	277.4	-11.8	37.1	219.5	81.7	5.9	1.018	0.913
S5	8	295.6	1.20	30.5	177.2	87.8	2.8	1.020	0.928
S6	11	295.1	-18.3	7.3	312.8	75.3	4.2	1.037	0.909
S7	27	344.2	8.1	9.9	121.0	82.5	1.5	1.022	0.888
S8	11	328.1	- 7.9	19.9	329.4	82.6	7.2	1.017	0.896
S9	31	5.6	- 9.4	6.9	21.0	76.2	2.6	1.032	0.901

* *Positive inclination indicates down.*

DISCUSSION OF MAGNETIC SUSCEPTIBILITY RESULTS

Stupavsky and Gravenor (1974) showed that the fine magnetite grains (<37 μ) are free to align themselves in the Earth's magnetic field to give the DRM, whereas the coarse grains (>37 μ) are not significantly aligned and do not contribute significantly to the DRM. The coarser magnetite grains either retain the orientation that was developed in the flowing ice, or they were oriented by the flow of the slurry at the ice-sediment interface. Thus, the orientation of the coarse magnetite grains causes the magnetic susceptibility anisotropy. Either of the above mechanisms provides a horizontal maximum susceptibility direction parallel to the ice-flow direction; the vertical minimum susceptibility direction is vertical.

The Leaside and Sunnybrook tills have nearly vertical minimum susceptibility directions (Fig. 7). All site mean maximum susceptibility directions for the Sunnybrook till are directed north and northwest and therefore agree with the ice-flow directions obtained by Karrow (1967) in pebble-orientation studies. In the case of the Leaside till, 1 site (L1) gave a northwest flow direction in agreement with Karrow's (1967) pebble studies. The remaining 5 sites, however, indicate a southwesterly flow direction. It should be pointed out that there are two Leaside tills, an Upper and a Lower, in the Toronto area (Dreimanis and Karrow, 1972). The samples referred to in this paper were taken from the Lower Leaside till. Fabric analyses on the Lower Leaside till (Dreimanis, oral commun.) are in agreement with the results reported in this paper.

Magnetic susceptibility (MS) measurements have been made on the Leaside and Sunnybrook till sheets (Gravenor and Stupavsky, 1974). The Leaside till has an MS value of $12.6 \pm 3.0 \times 10^{-5}$ cgs units, and the Sunnybrook till has a value of $7.38 \pm 0.92 \times 10^{-5}$ cgs units. The two populations are significantly different at the 99 percent confidence level as determined from both F and t tests. Therefore, MS is yet another magnetic property that can be used to differentiate the two till sheets.

CONCLUSIONS

Both the Leaside and Sunnybrook tills retain a stable primary magnetic remanence of detrital origin giving clearly distinguishable pole positions. This result confirms the conclusion of our earlier study (Gravenor and others, 1973) that till sheets can be characterized and statistically distinguished by their VGP position. Also, it confirms the hypothesis that these till sheets were deposited in a few hundred years or less.

The near-sided pole position of the Leaside till provides field evidence for the absence of inclination errors of more than a few degrees in the remanence of glacial tills. This is the same conclusion as that drawn by Stupavsky and Gravenor (1974) from their laboratory study.

In summary, the combination of magnetic properties—direction and intensity of remanent magnetization, anisotropy of magnetic susceptibility, and magnetic susceptibility—provide a unique signature for each till sheet. The potential of these methods in glacial studies appears to be promising.

ACKNOWLEDGMENTS

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